

Aengus McGuinness

(805) 570-1436 | amcguinness@college.harvard.edu | github.com/AengusMcGuinness | aengusmcguinness.com

EDUCATION

Harvard College

Cambridge, MA

A.B. Computer Science — GPA 3.88 / ACT 35, John Harvard Scholar

Expected May 2027

- **Completed:** Distributed Computing, Operating Systems, Modern Storage Systems, Computer Architecture, Systems Programming & Machine Organization, Computing Hardware, Machine Learning, Algorithms, Probability
- **Fall 2026:** Computing at Scale, Advanced Computer Networks, Advanced Topics in Computer Architecture, Systems Security, Modern AI Systems. Course Assistant for Computing Hardware (CS1410).

EXPERIENCE

Ironsight

San Francisco, CA

Member of Technical Staff, Infrastructure — on leave for senior year

June 2026 – Present

- Built and productionized the company's GPU inference platform — a self-hosted multimodal LLM on vLLM over autoscaled GCP spot fleets, ~10k lines of Terraform/Packer/Ansible — completing **61k+ production video inferences** in five weeks; pre-hydrated weight disks cut per-worker model load **8×** (7 min → 53 s).
- Colocated the inference engine inside each 8-GPU worker, deleting the load-balancer tier; gated the cutover on prompt-token parity against the live fleet and a crash drill proving exactly-once processing under spot preemption.
- Grew production ML throughput **3.3×** in five weeks (4.7k → 15.3k videos/week) while raising end-of-day pipeline completion from ~80% to **97–99.8%** on 60% higher daily intake.
- Designed cross-cloud data-residency enforcement (US-only vs. global) across **7 compute providers**: per-region queue pairs, fail-closed verification of provider-reported placement, and IAM session-policy denial — unblocking a major enterprise customer, zero placement violations at rollout.
- Hardened capacity acquisition and cost control: zonal fallback for spot stockouts (a 156-retry regional stall became first-try placement), metric-based idle-GPU detection bounding a ~\$1.3k/incident burn to ~15 min, MIG autohealing, and preemption alerting that absorbed **254 spot preemptions**.

Harvard Generative AI Research Program

Cambridge, MA

Research Fellow

June 2025 – Aug 2025

- Authored the SLURM automation toolkit for a GPU protein-structure refinement pipeline (OpenFold/ROCKET on A100s) — MSA generation and clustering, X-ray data prep, config-driven refinement sweeps, scoring — turning a hand-run multi-day workflow into reproducible one-command job submissions now used by the lab.

Los Alamos National Laboratory, B-GEN

Santa Fe, NM

Student Researcher

Summers 2022 – 2024

- Cut terabyte-scale genomics pipeline runtime ~80% via SLURM parallelization, memory tuning, and GPU offload of throughput-bound stages.

SYSTEMS & PERFORMANCE PROJECTS

RCache — One-Sided RDMA Distributed Cache | C++, libibverbs | [code](#)

Solo · Spring 2026

- Built a key-value cache with three transports over **libibverbs**: a TCP/RPC baseline, two-sided RDMA send/recv, and one-sided reads served straight out of a registered hash-table memory region, with an optional RDMA **FETCH_AND_ADD** maintaining LRU recency metadata without server CPU involvement.
- Across three CloudLab trials on Mellanox hardware, one-sided reads sustained **974k ops/s** at 8 clients with a flat **12 μs p99** — **13×** the throughput and **13×** lower tail latency than TCP (157 μs p99) — and isolated the metadata atomic's cost at 3–4 μs p99: recency tracking stays on the critical path even when reads leave it.

Chickadee OS Kernel | C++, x86-64

Spring 2026

- Implemented blocking system calls (**sys_waitpid**, **sys_msleep**) with a timer-driven sleep mechanism backed by a hashed timing wheel, removing busy-wait loops while preserving correctness under multi-core preemption.

Stream Buffer Prefetcher | C++, Intel Pin | [code](#)

Team of 4 · Spring 2026

- Built a Pin-based memory-hierarchy simulator implementing Jouppi's fixed-depth stream buffer; sweeping depth × stream count over SPEC CPU2006 showed **libquantum**'s effective L1D misses collapse **43×** while irregular workloads gain ~2% for up to **9×** the L2 traffic — the bandwidth-waste finding that motivated the team's adaptive-depth follow-on.
- Wrote the Pin harness and resume-on-kill parallel sweep driver (72 configs) used to evaluate that adaptive policy: **5.02×** speedup on **libquantum**, prefetch accuracy 78% → **83%** on **dealII**.

TECHNICAL SKILLS

Languages: C++, Rust, Python, Bash, SQL

Systems: Linux, concurrency, distributed systems, low-latency networking, RDMA (**libibverbs**), kernel development, Intel Pin

Infrastructure: AWS, GCP, Terraform/OpenTofu, Ansible, Docker Swarm, SLURM, vLLM, gRPC, Prometheus, Postgres